

# PROBLEM 4. EXCESS ENTALPY MODEL for ISOPROPANOL-WATER MIXTURE

A. For the narrow temperature range ( $0^{\circ}\text{C}$  to  $50^{\circ}\text{C}$ ) where the experimental data were taken thermodynamically consistent Excess property functions are given by polynomials of the form:  $M^E = x_1 x_2 (A_0 + A_1 x_1 + A_2 x_1^2 + \dots)$  where  $M$  is the thermodynamic property in our case Enthalpy. We assume that terms  $A_3 x_1^3 + \dots$  do not contribute significantly so the model for excess enthalpy can be written

$$H^E = x_1 x_2 (A_0 + A_1 x_1 + A_2 x_1^2) \text{ where}$$

$A_0, A_1, A_2$  constants which are functions of temperature must be determined from the experimental data

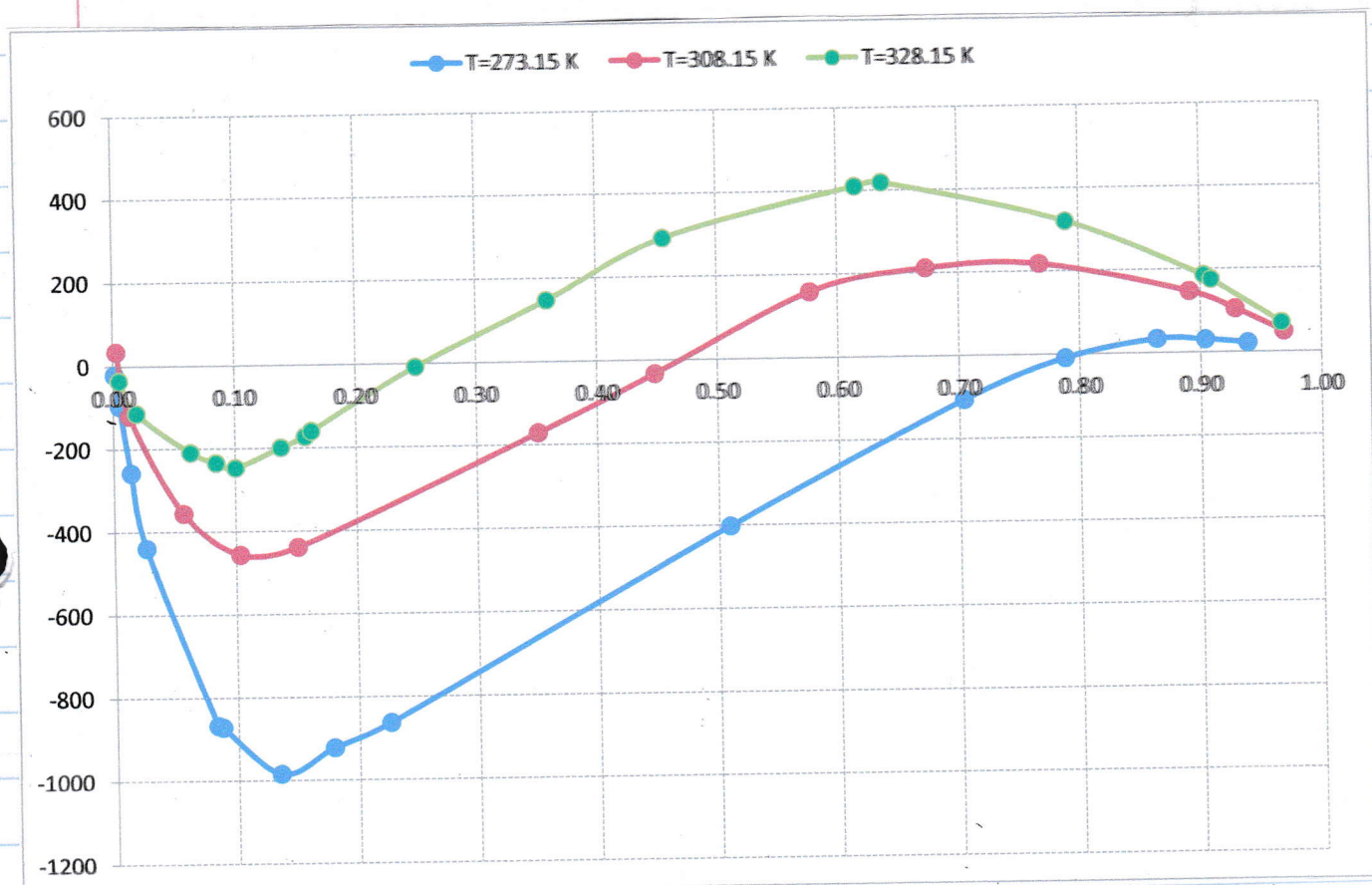
Performing multiple-linear Regression in Excel we get

| T      | $A_0$     | $A_1$    | $A_2$    |
|--------|-----------|----------|----------|
| 273.15 | -11900.20 | 30664.97 | -19744.7 |
| 308.15 | -6330.68  | 19380.44 | -12324.6 |
| 328.15 | -3572.27  | 15183.79 | -10306.3 |

To identify the effect of temperature, we observe that if we plot  $A_0, A_1, A_2$  vs.  $\ln T$  (or also vs  $T$ ) we obtain straight lines with very high linear correlation  $R^2 > 0.98$  for all coefficients

(2)

Below we see a graph of the excess enthalpy model against the experimental data at all three temperature levels.



The linear functional dependence of the coefficients  $A_0$ ,  $A_1$ ,  $A_2$  along with the final Excess Enthalpy Model are summarized here

$$A_0 = 45503 \ln T - 267150 \quad T [=] \text{ K}$$

$$A_1 = -85617 \ln T + 510,734$$

$$A_2 = 52796 \ln T - 315662$$

$$H^E = x_1 x_2 (A_0 + A_1 x_1^2 + A_2) \quad [= \text{ kJ/kmol }]$$



(3)

B. Now that we developed the model for Excess enthalpy we can calculate the molar enthalpy of any mixture using the formula

$$H = x_1 H_1 + x_2 H_2 + H^E \Rightarrow H = x_1 H_1 + x_2 H_2 + x_1 x_2 (A_0 + A_1 x_1 + A_2 x_1^2)$$

where  $H_1(P, T) = H_1^v(P^{sat}, T) - \Delta H_1^{vap}(T)$  and  
 $H_2(P, T) = H_2^v(P^{sat}, T) - \Delta H_2^{vap}(T)$

where  $H_1^v(P_1^{sat}, T)$ ,  $H_2^v(P_2^{sat}, T)$  is the enthalpy of vapor at temperature  $T$  and pressure  $P_i^{sat}$  where  $P_i^{sat}$  is the saturated vapor pressure of component  $i$

$\Delta H_1^{vap}(T)$  and  $\Delta H_2^{vap}(T)$  are the heats of vaporization of the two components

The saturated vapor pressures of the two components Isopropyl alcohol & water for temperatures  $< 50^\circ\text{C}$  are very low, so we can ignore the effect of pressure on liquid enthalpy and also assume that the vapor @  $P^{sat}$  behaves like an ideal gas.

Based on these assumptions

$$H_1(T) = H_1^{IG}(T) - \Delta H_1^{vap}(T) \quad (\text{Eq. 1})$$

$$H_2(T) = H_2^{IG}(T) - \Delta H_2^{vap}(T) \quad (\text{Eq. 2})$$

Let's now calculate total enthalpy before and after mixing. To do that we can use our calculator and (Eq. 1) & (Eq. 2) in page (3).

BEFORE MIXING  $H_1^{\text{tot}} = n_1 H_1$  where  $n_1 = 20$

$H_2^{\text{tot}} = n_2 H_2$   $n_2 = 80$

AFTER MIXING  $H^{\text{tot}} = n (x_1 H_1 + x_2 H_2 + H^E)$   $n = n_1 + n_2 = 100$

The heat released from the mixing @  $315.15^\circ\text{K}$  is then given by  $Q = H^{\text{tot}} - H_1^{\text{tot}} - H_2^{\text{tot}} \Rightarrow$

$$Q = n(x_1 H_1 + x_2 H_2 + H^E) - n_1 H_1 - n_2 H_2 \Rightarrow \boxed{Q = n H^E}$$

To calculate  $H^E$  @  $315.15^\circ\text{K}$  first we calculate

$$A_0 = 45503 \ln(315.15) - 267150 \Rightarrow A_0 = -5369.0$$

$$A_1 = -85617 \ln(315.15) + 510,734 \Rightarrow A_1 = 18175.2$$

$$A_2 = 52796 \ln(315.15) + 315,662 \Rightarrow A_2 = -11924.0$$

Using these values we obtain  $H^E = -353.8 \text{ kJ/kmol}$  and therefore the heat released

$$Q = n H^E = 0.1 \cdot (-353.8) \Rightarrow$$

$$\boxed{Q = -35.38 \text{ kJ}}$$



- c. For adiabatic mixing we have that the total enthalpy before is the same as the total enthalpy after the mixing.

Let's assume that the temperature after the mixing is equal to  $T_m$ , and before the mixing is  $T$ . Then from the energy balance we have

$$n_1 H_1(T) + n_2 H_2(T) = n (x_1 H_1(T_m) + x_2 H_2(T_m) + x_1 x_2 (A_0(T_m) + A_1(T_m)x_1 + A_2(T_m)x_1^2))$$

This equation must be solved for  $T_m$ . Dividing left and right hand side by  $n$  we obtain

$$x_1 H_1(T) + x_2 H_2(T) = x_1 H_1(T_m) + x_2 H_2(T_m) + x_1 x_2 (A_0(T_m) + A_1(T_m)x_1 + A_2(T_m)x_1^2)$$

The extended secant method was used with results shown here

|                      |             |            |            |           |              |
|----------------------|-------------|------------|------------|-----------|--------------|
| $A_0$                | -4938.6377  | $x_1$      | 0.2000     |           |              |
| $A_1$                | 17365.4263  | $x_2$      | 0.8000     |           |              |
| $A_2$                | -11424.6731 |            |            |           |              |
|                      |             |            |            |           |              |
| <b>Before Mixing</b> |             | $H_1$      | $H_2$      | $H^E$     | <b>Total</b> |
| Temperature          | 42.0000     | -314520.72 | -285142.93 | 0.0000    | -291018.49   |
|                      |             |            |            |           |              |
| <b>After Mixing</b>  |             | $H_1$      | $H_2$      | $H^E$     | <b>Total</b> |
| Temperature          | 44.9950     | -313980.19 | -284893.55 | -307.6063 | -291018.49   |

The temperature of the solution after the mixing rises by approximately  $3^\circ\text{C}$  to around  $45^\circ\text{C}$